

IoT Device Provisioning Policy

The Grand Marina

Prepared by: Hydroficient Engineering

Version 1.0 | Effective Date: 2025

1. Purpose

This policy governs the provisioning, certificate lifecycle management, and decommissioning of all HYDROLOGIC IoT devices deployed at The Grand Marina. It ensures every device connecting to the Hydroficient MQTT infrastructure has a verifiable identity, that credentials are managed securely throughout their lifespan, and that unauthorized devices cannot gain network access.

This document applies to all Hydroficient technicians, Grand Marina facilities staff, and any third-party contractors involved in IoT device installation or maintenance.

2. Scope and Device Inventory

This policy covers all HYDROLOGIC devices installed at The Grand Marina:

Device ID	Location	Certificate	Expires	Status
HYDROLOGIC-MainBuilding-001	Mechanical Room A	device-001.pem	Feb 2027	Active
HYDROLOGIC-PoolSpa-002	Pool Pump House	device-002.pem	Feb 2027	Active
HYDROLOGIC-Kitchen-003	Kitchen Mechanical	device-003.pem	Feb 2027	Active

3. Roles and Responsibilities

3.1 Hydroficient technicians

- Generate CSRs and submit them to the CA for signing
- Install signed certificates and configure devices before deployment
- Log all provisioning events with device ID, date, and technician name

3.2 Hydroficient project lead

- Approve all provisioning and decommissioning requests

- Maintain custody of the CA private key in AWS Secrets Manager
- Authorize certificate revocations and confirm revocation is recorded in the CRL

3.3 Grand Marina facilities manager

- Report lost, stolen, or physically damaged devices to Hydroficient immediately
- Confirm device locations match the inventory table in Section 2 on an annual basis

4. Device Onboarding

Every new HYDROLOGIC device must complete all steps below before connecting to the production MQTT broker. No device may go live without a certificate signed by the Hydroficient CA.

4.1 Onboarding procedure

1. Generate a CSR on the device. The private key must be generated on-device and must never leave the device.
2. Submit the CSR to the Hydroficient CA operator for signing.
3. Install the signed certificate and the CA bundle on the device.
4. Register the device ID and certificate thumbprint in AWS IoT Core.
5. Configure the MQTT broker endpoint and topic permissions on the device.
6. Verify a successful mTLS handshake with the broker.
7. Confirm the device publishes telemetry to the correct topic.
8. Record the provisioning date, device ID, location, and technician name in the device log.

4.2 Key storage requirements

- Device private keys: generated and stored on-device only, never transmitted or backed up
- CA private key: stored in AWS Secrets Manager, IAM access restricted to the project lead role
- Certificate files: stored in the Hydroficient device registry with access controls

5. Certificate Lifecycle Management

All device certificates follow a six-stage lifecycle:

Stage	When	Who acts	What happens
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1. CSR generation	New device onboarding	Technician	CSR generated on-device; private key stays on device
2. Validation	On CSR submission	Project lead	Identity and authorization verified before signing
3. Issuance	After validation	CA operator	Certificate signed with 1-year validity
4. Installation	Before deployment	Technician	Cert + CA bundle installed; mTLS verified
5. Monitoring	60 days before expiry	Automated alert	Renewal window opens; technician schedules update
6. Renewal / Revocation	At expiry or compromise	Project lead + technician	New cert issued, or cert revoked immediately

5.1 Certificate renewal

- Renewal alerts fire 60 days before expiry
- A new CSR is generated and a new certificate is issued (e.g., device-001-v2.pem)
- The device is updated during a scheduled maintenance window
- The old certificate is revoked immediately after the new one is confirmed working
- Renewal must be completed no later than 7 days before the expiry date

6. Breach and Compromise Response

If a device is lost, stolen, or suspected of compromise, its certificate must be revoked immediately. Do not wait for the next scheduled maintenance window.

6.1 Immediate revocation procedure

9. Grand Marina facilities manager reports the incident to the Hydroficient project lead.
10. Project lead revokes the device certificate in AWS IoT Core within 2 hours of notification.
11. Revocation is recorded in the Certificate Revocation List (CRL).
12. The broker is confirmed to be rejecting connections from the revoked certificate.
13. An incident report is filed documenting the timeline, device ID, and actions taken.
14. If the device is recovered, it must be fully re-provisioned with a new certificate before returning to service. The compromised certificate is permanently blacklisted.

Critical: If the CA private key is suspected of exposure, all certificates signed by that CA must be revoked and re-issued under a new CA.

7. Device Decommissioning

When a HYDROLOGIC device is permanently removed from service, its certificate must be revoked on the same day it goes offline.

15. Grand Marina facilities manager submits a decommission request to Hydroficient.
16. Project lead approves and schedules the decommission.
17. Technician physically removes the device and confirms it is offline.
18. Project lead revokes the certificate in AWS IoT Core.
19. Certificate is added to the permanent blacklist and may not be reinstated.
20. Device is returned to Hydroficient for secure disposal or refurbishment.
21. Device registry is updated to reflect decommissioned status.

8. Adding Devices During Site Expansion

When The Grand Marina expands, additional HYDROLOGIC devices are provisioned following Section 4. Each new device receives its own unique certificate signed by the Hydroficient CA. The device inventory table in Section 2 must be updated before any new device goes live.

For deployments of 10 or more devices simultaneously, Hydroficient will use automated provisioning via AWS IoT Core fleet provisioning to reduce manual error and ensure consistent configuration.

9. Policy Review

- This policy is reviewed annually or after any security incident
- The Hydroficient project lead is responsible for maintaining this document
- All changes must be documented in the version history below

Version	Date	Author	Changes
1.0	2025	Hydroficient Engineering	Initial release